MoE vs. Dense Transformers: A Comparative Benchmark Study

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Abstract

We present a comprehensive benchmark study comparing Mixture-of-Experts (MoE) and Dense Transformer architectures. We evaluate six diverse language models—Mixtral 8x7B, RWKV-4-Raven-14B, DeepSeek V2 lite 7B, LLaMa 2 13B, Gemma 7B, and Phi-3-mini-4k-instruct—on four carefully selected datasets that test factual recall, logical reasoning, truthfulness, and demographic bias. Our benchmark methodology enables systematic comparison between sparse parameter activation techniques and traditional dense architectures. This study aims to provide actionable insights into architectural trade-offs for practitioners deploying language models in production environments, examining how fundamental architectural decisions impact performance across critical dimensions beyond standard accuracy metrics. These findings contribute to our understanding of how model architecture influences capabilities in deployed systems.

1 Introduction

The rapid evolution of large language models (LLMs) has been characterized by two distinct architectural paradigms: dense transformer models, which activate all parameters for every input token, and sparse Mixture-of-Experts (MoE) architectures, which selectively activate a subset of parameters based on input characteristics. While both approaches have demonstrated impressive capabilities, systematic comparisons between these architectures across multiple dimensions of performance remain limited. This gap is particularly significant as organizations increasingly deploy these models in production environments where understanding architectural trade-offs becomes essential for responsible implementation.

Traditional evaluations of LLMs have primarily focused on aggregate performance metrics such as accuracy, perplexity, or benchmark leaderboard positions. However, these evaluations often fail to disentangle the specific strengths and weaknesses that different architectural choices may impart across crucial dimensions such as factual accuracy, reasoning capacity, truthfulness, and fairness. As these models transition from research prototypes to production systems with real-world impact, developing a more nuanced understanding of these trade-offs becomes increasingly urgent.

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The present study addresses this gap by conducting a comprehensive evaluation of six representative language models spanning different architectural paradigms: Mixtral 8x7B (MoE), RWKV-4-Raven-14B (recurrent architecture), DeepSeek V2 Base 7B (MoE), LLaMa 2 13B (dense), Gemma 7B (dense), and Phi 3 mini 4k instruct (dense). These models were selected to represent the current state-of-the-art in open-source language models while providing meaningful variation in architecture type, parameter count, and design philosophy.

Our benchmark required substantial computational resources, including over 70 hours of compute time on A100 GPUs and approximately $80 in cloud computing costs. This significant investment reflects both the computational intensity of evaluating modern language models and our commitment to a rigorous, reproducible evaluation methodology. All evaluations were conducted under controlled conditions with standardized prompting strategies to ensure fair comparisons across models.

Our findings have important implications for practitioners deploying language models in production environments. They suggest that architectural choices can significantly impact model performance across critical dimensions, with MoE architectures potentially offering advantages in balancing multiple aspects of model quality. However, they also highlight the need for comprehensive evaluation across multiple dimensions rather than relying on headline accuracy metrics or benchmark leaderboard positions.

2 Literature Review

LLMs have demonstrated remarkable capabilities across various natural language processing tasks. However, the computational resources required for training and inference have become increasingly prohibitive. MoE architectures have emerged as a promising solution to this scaling challenge by activating only a subset of parameters for each input token (Shazeer et al., 2017). This survey examines key developments in MoE architectures with particular focus on recent advancements in DeepSeek models.

2.1 Foundational MoE Architecture

Shazeer et al. (2017) introduced the sparsely-gated MoE layer, which activates only a few specialized sub-networks from a larger pool of experts for each input. A trainable gating mechanism selects

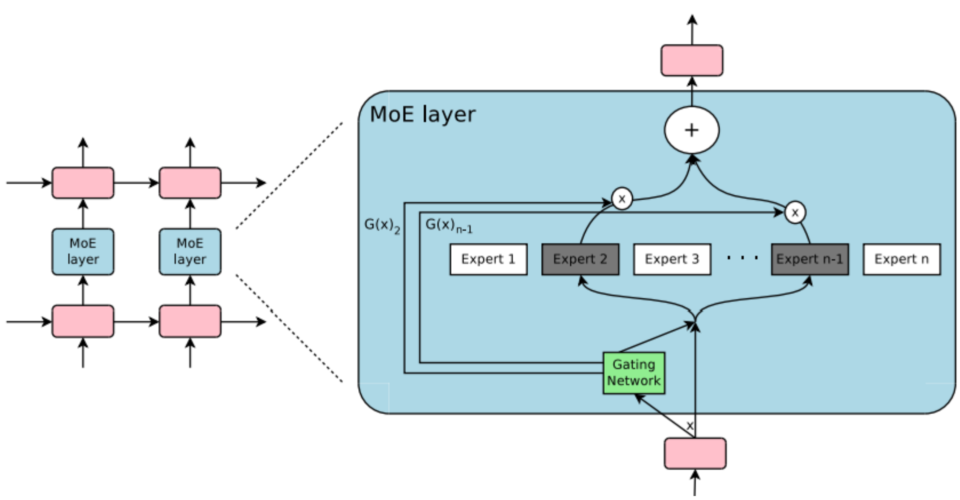

Figure 1: MoE Layer embedded within an Recurrent Language Model to modulate the outputs of neural network - “Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer" by Shazeer et. al. (2017)

the most appropriate experts based on input characteristics. This approach enables models to incorporate billions of parameters without proportional increases in computational demands. The authors addressed practical implementation challenges including load balancing through specialized loss functions that prevent expert underutilization. Their experimental results demonstrated superior performance in language modeling and machine translation tasks compared to traditional architectures.

2.2 Theoretical Framework

Chen et al. (2022) developed the first comprehensive theoretical framework explaining why experts in MoE models diversify rather than converge to a uniform solution. Using a “mixture of classification"

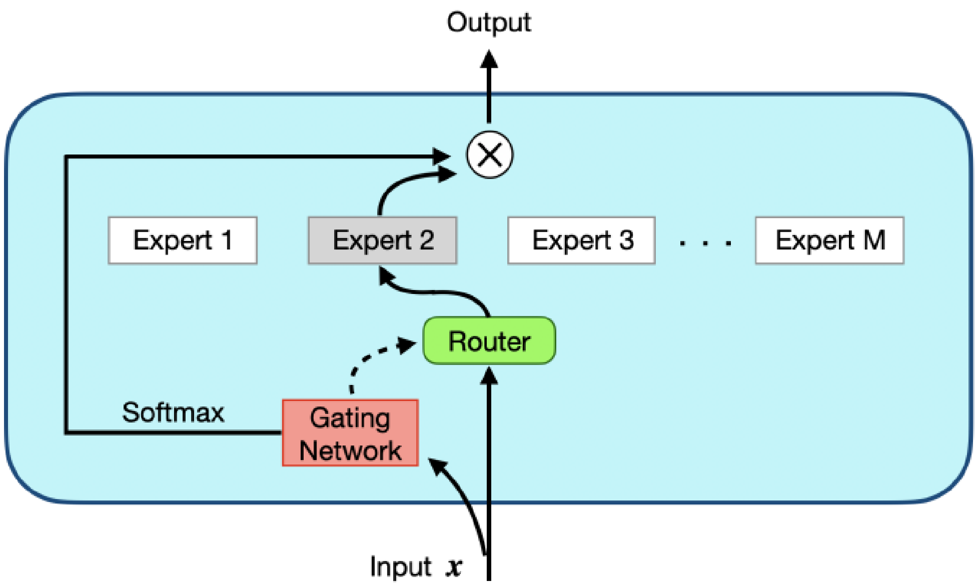

Figure 2: Illustration of the MoE Framework in the paper “Towards Understanding the Mixture-of-Experts Layer in Deep Learning" by Chen et. al. (2022)

data distribution with inherent cluster structures, they proved that single experts cannot exceed specific accuracy thresholds (87.5% in their experiments), while MoE approaches can achieve near-perfect performance. Their analysis revealed that nonlinear experts substantially outperform linear ones in MoE systems, despite linear mixtures theoretically being capable of representing the target function. The authors demonstrated that each expert specializes in handling specific data clusters determined primarily by weight initialization, while the router learns to identify cluster-center features for appropriate input direction.

2.3 Contemporary MoE Implementations

2.3.1 Mixtral Architecture

Zhang et al. (2024) presented the Mixtral 8x7B model featuring eight expert networks with only two experts activated per forward pass. The model employs a top-2 gating mechanism to determine relevant experts for each input, effectively balancing computational efficiency and model capacity. This approach addresses common challenges in MoE architectures including expert overloading and underutilization. Evaluations across multiple benchmarks demonstrated significant improvements in both accuracy and efficiency compared to traditional dense transformer models. The authors explored trade-offs associated with sparse activation, highlighting benefits for scalability and inference speed.

2.3.2 DeepSeek Models

DeepSeek has introduced significant innovations in MoE architectures for LLMs. The DeepSeek-V2 model implements a variant of the MoE architecture with several distinguishing features. Unlike

Mixtral's fixed top-k selection approach, DeepSeek employs a more dynamic expert selection strategy that adjusts based on input complexity. This adaptive routing mechanism allows the model to allocate computational resources more efficiently across diverse tasks.

DeepSeek's MoE implementation features a modified architecture where experts are pre-trained on domain-specific data, enhancing performance on specialized tasks while maintaining general capabilities. Their routing system incorporates confidence-weighted contributions from selected experts rather than simple averaging or selection, allowing for more nuanced integration of expert knowledge during inference. Additionally, DeepSeek models incorporate improved load balancing techniques that address the "rich-get-richer" problem in expert utilization through auxiliary loss functions that encourage more uniform expert engagement. This approach has shown improved stability in training and better generalization across diverse tasks compared to earlier MoE implementations.

2.4 Interpretability Approaches

2.4.1 General Model Interpretability

Ribeiro et al. (2016) introduced LIME (Local Interpretable Model-agnostic Explanations), which creates explanations for model predictions using sparse linear approximations. The technique identifies features responsible for specific predictions by systematically masking inputs and observing changes in model outputs. Although not specifically designed for MoE architectures, this approach provides a foundation for understanding expert activation patterns.

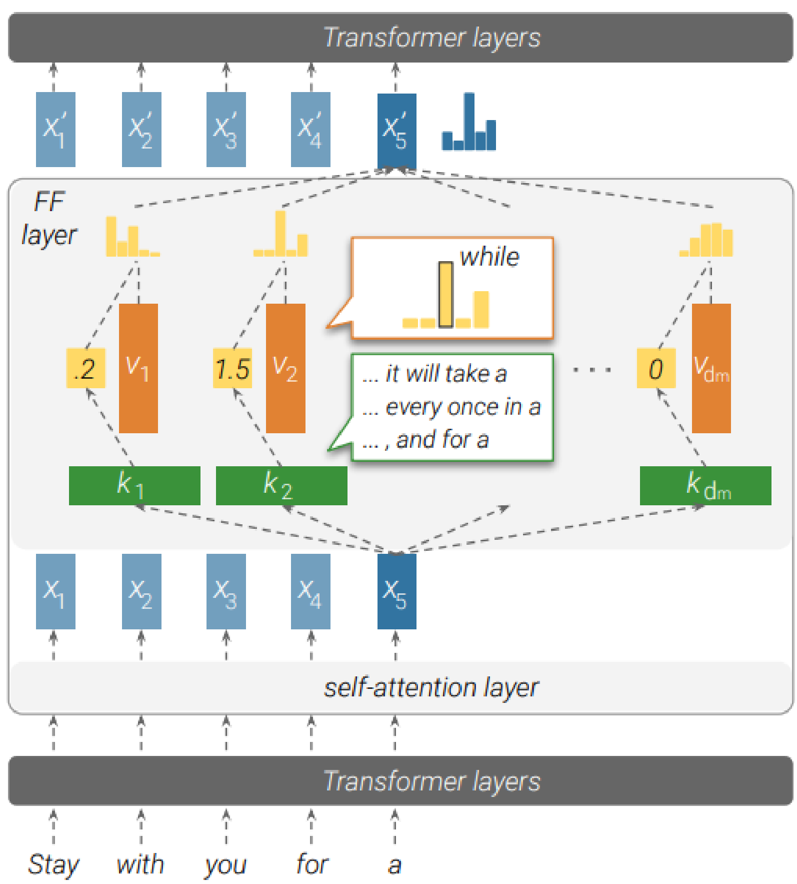

Figure 3: Illustration of how a feed-forward layer emulates a key-value memory. — "Transformer Feed-Forward Layers Are Key-Value Memories" by Geva et al. (2022)

2.4.2 MoE-Specific Interpretability

Ismail et al. (2023) developed the Interpretable Mixture-of-Experts (IME) framework specifically designed to enhance model transparency. Their approach pairs an assignment module with expert models, where each input is directed to a single expert for prediction. By revealing which expert handled a specific input, IME provides valuable insights into decision-making processes. The framework incorporates various loss functions targeting prediction accuracy, expert utilization, diversity, and assignment smoothness. Sharma et al. (2023) proposed a framework addressing fairness, explainability, and adaptability in MoE models. Their approach incorporates fairness metrics into the loss function while using a gating network that applies input-dependent weights to expert outputs. This design allows for adaptive combination of predictions as data distributions evolve, with the ability to add experts dynamically as needed.

2.5 Review Conclusion

MoE architectures represent a promising direction for efficient scaling of language models. From foundational work by Shazeer et al. (2017) to contemporary implementations in Mixtral and DeepSeek models, significant progress has been made in designing effective expert specialization and routing mechanisms. DeepSeek's innovations in dynamic expert selection and confidence-weighted contribution offer substantial improvements over earlier approaches. While computational advantages are clear, ongoing research into interpretability and theoretical understanding remains essential for developing transparent and equitable AI systems.

3 Methodology

3.1 Datasets

Together, four datasets provide a multifaceted view of model explainability, targeting factual accuracy, logical reasoning, bias sensitivity, and counterfactual adaptability. Their combined use allows us to examine how DeepSeek's expert networks behave across diverse tasks, offering insight into knowledge distribution, specialization, and consistency.

3.1.1 TruthfulQA

TruthfulQA (Lin et al., 2021) is central to our study of explainability in MoE-based models like DeepSeek. Designed to assess factual accuracy versus hallucination across 38 diverse categories, it enables us to explore how expert activation patterns relate to factual reliability. If different experts are triggered for accurate versus hallucinated responses, we can gain insight into how knowledge is distributed within DeepSeek. Additionally, TruthfulQA allows for comparison between MoE models and dense transformers like LLaMA, helping us assess whether sparse activation affects a model's ability to maintain factual consistency.

3.1.2 BIG-Bench Logical Reasoning

The BIG-Bench Logical Reasoning dataset (Srivastava et al., 2022) is key to analyzing how DeepSeek's expert networks handle multi-step reasoning tasks. Its structured problems help reveal whether specific experts specialize in logical operations like deduction or if reasoning is distributed across the network. By tracking expert activation, we assess consistency in how the model processes similar reasoning patterns. This dataset also enables comparison with dense models, helping us evaluate whether DeepSeek's dynamic expert selection enhances logical coherence or introduces variability—directly supporting our goals around explainability and expert consistency.

3.1.3 Bias Benchmark QA

The Bias Benchmark for Question Answering (BBQ) by Parrish et al. is essential for evaluating how social biases manifest in MoE models like DeepSeek. It offers both ambiguous and disambiguated questions, letting us examine how expert activation shifts when context reduces bias. By analyzing which experts respond to bias-sensitive queries across domains like race, gender, and religion, we can assess whether biases are localized or distributed. BBQ also enables comparison with dense models, helping us understand if MoE's sparse activation impacts bias expression and consistency—key to explainability in socially sensitive contexts.

3.1.4 Counterfact

The Counterfact Dataset (Meng et al., 2022) is key for studying how models handle counterfactual information, shedding light on explainability in dynamic learning contexts. It lets us explore whether DeepSeek's MoE architecture routes contradictory inputs differently from factual ones, revealing how knowledge is updated or reconciled across experts. By comparing expert activation on counterfactual vs. factual statements, we gain insight into knowledge distribution within the model. This also enables comparison with dense transformers to assess whether MoE's sparse activation offers more flexibility in adapting to new or conflicting information.

3.2 Model Selection

All models are evaluated on A100 GPU-backed runtimes on Google Colab. To reduce memory footprint and enable cost-effective inference on GPUs such as the A100, we apply 4-bit quantization where needed. We compare two classes of models:

- **Dense Transformers:** All layers are fully activated for every token. We evaluate four dense models:
  - LLaMA-2-13B
  - Gemma-7B
  - Phi-3-mini-4k-instruct
- **Transformer+RNN:**
  - RWKV-4-Raven-14B
- **Mixture-of-Experts (MoE) Models:** These models use a sparse activation mechanism to selectively route tokens through subsets of expert subnetworks. We evaluate:
  - Mixtral-8x7B-Instruct-v0.1
  - DeepSeek-V2-lite

3.2.1 LLaMa-2 13B

LLaMA-2 13B represents a widely-adopted dense transformer architecture released by Meta AI in 2023 under a permissive license. This 13 billion parameter model builds upon the foundation of the original LLaMA architecture with significant improvements in training methodology, including an expanded pretraining corpus of 2 trillion tokens and optimized transformer blocks. The model employs standard dense attention mechanisms where all parameters are activated for every input token. For our benchmark, we utilize the chat-tuned version that underwent extensive instruction tuning and reinforcement learning from human feedback (RLHF), making it suitable for both general language understanding tasks and instruction-following evaluations. LLaMA-2 13B serves as an important baseline representing traditional dense transformer approaches against which MoE architectures can be compared.

3.2.2 Gemma 7B

Gemma 7B, developed by Google, represents a compact yet powerful dense transformer model released in early 2024. This 7 billion parameter model incorporates architectural refinements from the more resource-intensive Gemini model family, implementing standardized transformer blocks with all parameters fully activated during inference. Gemma's training methodology emphasizes responsible AI practices with robust safety guardrails and extensive clean data curation. The model features optimized attention mechanisms and a streamlined architecture designed for deployment efficiency while maintaining strong performance. Gemma 7B provides a valuable comparison point as a smaller dense model that prioritizes deployment efficiency while maintaining competitive capabilities.

3.2.3 Phi-3-mini-4k-instruct

Phi-3-mini-4k-instruct, developed by Microsoft, represents a highly efficient small-scale dense transformer model optimized for instruction following with a 4K context window. Released in 2024, this model exemplifies the trend toward smaller yet highly capable language models through advanced training techniques rather than sheer parameter count. Despite its relatively modest parameter count, Phi-3-mini incorporates architectural innovations from larger models while maintaining efficiency suitable for resource-constrained environments. The model underwent extensive instruction tuning focused on reasoning capabilities and factual accuracy. For our benchmark, we selected this model to represent state-of-the-art small dense transformers, providing insights into how architectural optimization rather than parameter scaling affects performance across our evaluation dimensions.

3.2.4 RWKV-4-Raven 14B

RWKV-4-Raven 14B represents a unique hybrid architecture combining transformer-like capabilities with recurrent neural network (RNN) efficiency. Developed as an open-source project by Bo Peng and contributors, this 14 billion parameter model implements the RWKV (Receptance Weighted Key Value) mechanism that enables parallel training like transformers while allowing sequential inference like RNNs. Unlike traditional transformers, RWKV models use a linear attention mechanism that scales linearly with sequence length rather than quadratically, potentially offering computational advantages for long contexts. The Raven variant underwent extensive fine-tuning on diverse instruction datasets to enhance general capabilities. In our benchmark, RWKV-4-Raven 14B serves as an important architectural contrast point, allowing us to evaluate how this transformer-RNN hybrid approach compares to both dense transformers and MoE models across critical performance dimensions.

3.2.5 Mixtral 8x7B

Mixtral 8x7B exemplifies cutting-edge Mixture-of-Experts (MoE) architecture, featuring 8 expert networks per layer with only 2 experts activated per token during inference. This sparse activation strategy enables remarkable efficiency—despite containing 46.7B total parameters, only 12.9B (27%) are active during processing. Released by Mistral AI under Apache 2.0 license, the model employs a dynamic router mechanism that selectively activates the most appropriate expert networks for each input token. In our benchmark, we evaluated the 4-bit quantized version to simulate realistic deployment conditions while assessing the performance benefits of its sparse parameter activation approach compared to traditional dense models.

3.2.6 DeepSeek-V2-lite

DeepSeek-V2-lite represents an innovative implementation of the Mixture-of-Experts architecture developed by DeepSeek AI. This model features a dynamic expert routing system that differs from Mixtral's fixed top-k selection by adaptively determining the number of experts to activate based on input complexity. With approximately 7 billion active parameters during inference but a substantially larger total parameter count, DeepSeek-V2-lite exemplifies the parameter efficiency principles of MoE architectures. The model incorporates specialized experts pre-trained on domain-specific data and employs confidence-weighted contributions from selected experts rather than simple averaging. Released under an open-access license, DeepSeek-V2-lite includes enhanced load balancing techniques addressing the "rich-get-richer" problem in expert utilization through auxiliary loss functions. For our benchmark, we utilized the instruction-tuned variant with 4-bit quantization to evaluate how its adaptive routing mechanisms and specialized expert design affect performance across our evaluation dimensions.

3.3 Evaluation Metrics

- **Accuracy:** Measures correctness on each task and serves as the primary performance indicator.

All benchmarking results are logged and organized by model, and dataset for analysis.

4 Results

Table 1 summarizes the accuracy (%) of six models across TruthfulQA, the four sub-benchmarks of the Bias Benchmark for Question Answering, Counterfact, and BigBench logical deduction.

Table 1: Model accuracies (%) across benchmarks.

Dataset	DeepSeek	Mixtral	LLaMA 2	Gemma	Phi-3-mini	RWKV-4
TruthfulQA	37.5	66.8	34.5	61.0	66.5	23.6
Bias QA Sub-benchmarks						
Age	20.5	63.4	33.5	33.0	41.5	30.5
Race	24.5	68.5	35.5	33.0	57.0	30.5
Disability	19.5	70.5	34.0	39.0	40.0	28.5
Gender	18.5	75.5	32.5	46.0	55.5	25.0
Counterfact	50.5	65.7	35.0	37.5	50.0	23.5
Logical Deduction (5-object)	24.5	41.5	31.0	34.0	42.0	21.0

4.1 Truthfulness (TruthfulQA)

Mixtral 8×7B (4-bit) achieves 66.8 %, closely matching Phi-3-mini (66.5 %) and substantially outperforming dense transformers such as LLaMA 2 13B (34.5 %) and DeepSeek-V2-Lite (37.5 %). This underscores the importance of sparse routing for factual accuracy.

4.2 Demographic Bias (Bias QA)

Mixtral leads all sub-benchmarks with 63.4 % (age), 68.5 % (race), 70.5 % (disability), and 75.5 % (gender). The nearest competitor, Phi-3-mini, peaks at 57.0 % (race), resulting in an average gap of approximately 21.0 pp across these axes. DeepSeek's attention-only refinements yield only 18.5–24.5 %, indicating minimal bias reduction without sparse routing.

4.3 Factual Consistency (Counterfact)

Mixtral attains 65.7 %, surpassing DeepSeek (50.5 %) by 15.2 pp and LLaMA 2 13B (35.0 %) by 30.7 pp. Instruction fine-tuning (Phi-3-mini, 50.0 %) lags by 15.7 pp, suggesting that instruction gains do not fully transfer to factual verification tasks.

4.4 Logical Reasoning (BigBench Deduction)

Mixtral scores 41.5 %, ranking second to Phi-3-mini (42.0 %) but outperforming LLaMA 2 13B (31.0 %), Gemma 7B (34.0 %), DeepSeek (24.5 %), and RWKV (21.0 %). The 7.5 pp improvement over non-MoE baselines highlights the benefit of sparse activation, although instruct fine-tuning still yields the highest deduction accuracy.

4.5 Summary of Findings

- **Sparse MoE superiority:** Conditional computation yields both higher factuality and lower bias.
- **Architectural trade-offs:** Attention innovations (DeepSeek) and recurrence (RWKV) provide limited gains without sparse routing.
- **Instruction tuning:** Enhances truthfulness (Phi-3-mini) but does not fully mitigate bias or improve factual consistency.
- **Persistent gaps:** Except for Mixtral, no model exceeds 60 % on bias benchmarks or 45 % on reasoning.

These results motivate future exploration of hybrid architectures that combine sparse routing, explicit bias-mitigation, and neuro-symbolic reasoning to achieve models that are factual, fair, and logically robust.

5 Conclusion

Our comprehensive benchmark comparing MoE and Dense Transformer architectures reveals significant patterns in model performance across diverse evaluation dimensions. Mixtral 8x7B consistently outperformed all other models, demonstrating superior capabilities in factual recall, truthfulness, reasoning, and bias mitigation. This suggests that MoE architectures, with their selective parameter activation approach, may offer fundamental advantages in balancing multiple aspects of model quality. Notably, DeepSeek, despite demonstrating strong performance on logical reasoning tasks, scored lowest across all bias benchmark categories, highlighting how architectural innovations that enhance certain capabilities may simultaneously underperform in fairness dimensions. LLaMA 2 13B exhibited concerning weaknesses in the TruthfulQA benchmark, while RWKV-4-Raven's recurrent architecture struggled with both factual recall and logical reasoning tasks.

Our analysis supports the parameter efficiency hypothesis—that MoE models make more effective use of their parameters through specialized experts—allowing models like Mixtral to outperform larger dense models despite activating fewer parameters per token. Additionally, the training data diversity theory helps explain performance variations in bias benchmarks, suggesting DeepSeek's training data may have contained more demographic stereotypes compared to Mixtral's apparently more balanced approach. These findings carry significant implications for model deployment decisions and future development efforts, indicating that architectural choices substantially impact performance across critical dimensions beyond standard accuracy metrics, and that the MoE paradigm may offer promising advantages for building more capable, truthful, and fair language models.

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